

```
import cv2
import numpy as np
from google.colab import files
from matplotlib import pyplot as plt

# Upload image
uploaded = files.upload()

# Get uploaded image name
filename = list(uploaded.keys())[0]

# Read image in grayscale
img = cv2.imread(filename, 0)

# Check if image was loaded
if img is None:
    print("Error: Image could not be loaded.")
else:
    # Apply Sobel X edge detection
    sobel_x = cv2.Sobel(img, cv2.CV_64F, 1, 0, ksize=5)

    # Convert result to uint8
    sobel_x = cv2.convertScaleAbs(sobel_x)

    # Save output
    cv2.imwrite("sobel_x.jpg", sobel_x)

    # Display input and output
    plt.figure(figsize=(10, 4))

    plt.subplot(1, 2, 1)
    plt.imshow(img, cmap="gray")
    plt.title("Original Image")
    plt.axis("off")

    plt.subplot(1, 2, 2)
    plt.imshow(sobel_x, cmap="gray")
```

```
plt.title("Sobel X Edge Detection")
plt.axis("off")

plt.show()

# Download output image
files.download("sobel_x.jpg")
```

[Choose Files](#) anime-boy-...ace-sign.jpg

anime-boy-giving-peace-sign.jpg(image/jpeg) - 9355745 bytes, last modified: 8/19/2026 - 100% done

Saving anime-boy-giving-peace-sign.jpg to anime-boy-giving-peace-sign.jpg

Original Image



Sobel X Edge Detection



